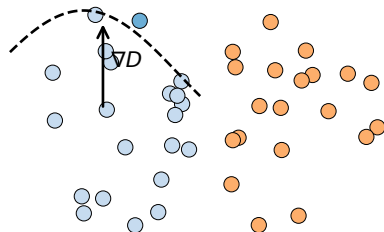


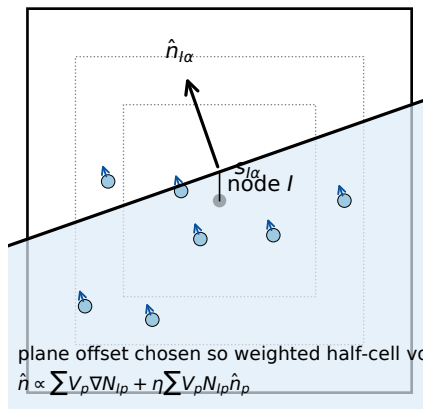
(a) Velocity fields from groups and DFG files (b) Implicit volume gradient plus optional explicit influence

$N_g = 2$ groups, $c \in \{0, 1\}$ DFG flag
group 0, DFG side 0 group 1



group 0, DFG side 1

field index: $\alpha = cN_g + g$



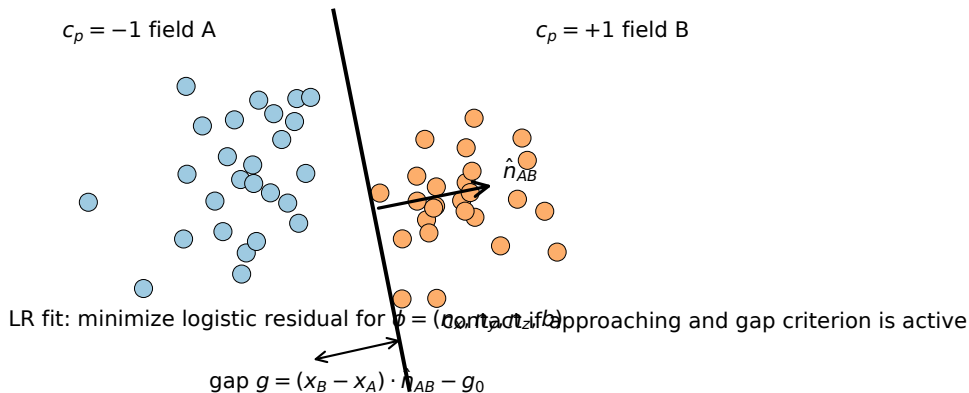
plane offset chosen so weighted half-cell volume matches $V_{I\alpha}$

$$\hat{n} \propto \sum V_p \nabla N_{Ip} + \eta \sum V_p N_{Ip} \hat{n}_p$$

(c) Logistic-regression normal and normal-gap contact law

$c_p = -1$ field A

$c_p = +1$ field B



LR fit: minimize logistic residual for $\phi = (n \cdot (x_B - x_A))$

$$\text{gap } g = (x_B - x_A) \cdot \hat{n}_{AB} - g_0$$